VIDEO EMOTION RECOGNITION USING MELD DATASET

CHAPTER 1

1.1 Introduction

Emotion recognition is an important research area in Artificial Intelligence and Affective Computing that enables machines to understand human emotional states. Human emotions are expressed through facial expressions, eye movements, eyebrow movements, lip movements, and facial muscle activities. The automatic identification of emotions from video sequences helps intelligent systems interact more naturally with humans.

Video Emotion Recognition (VER) aims to identify emotional states by analyzing facial expressions present in video data. Recent developments in Computer Vision, Deep Learning, and Transformer architectures have significantly improved the performance of emotion recognition systems. Applications include healthcare monitoring, intelligent tutoring systems, human-computer interaction, surveillance systems, virtual assistants, and behavioral analysis.

However, recognizing emotions from videos remains challenging due to variations in facial appearance, illumination conditions, head movements, occlusions, and dynamic emotional transitions.

1.2 Problem Statement

Most existing emotion recognition systems rely on multimodal information such as audio, text, and video. However, video-based emotional cues alone contain rich information through facial expressions and facial landmark movements.

The major challenges include:

- Variations in facial expressions among individuals.
- Similarity between different emotions.
- Dynamic changes in emotions across video frames.
- Class imbalance in emotion datasets.
- Difficulty in capturing temporal emotional dependencies.

Therefore, an efficient video emotion recognition framework is required to accurately classify emotions using only visual information.

1.3 Objectives

The objectives of this project are:

- To develop a video-based emotion recognition system using the MELD dataset.
- To extract facial landmarks and eyebrow movement features.
- To learn spatial facial representations using CNN.
- To learn temporal emotional dependencies using BiLSTM.
- To improve classification performance using Attention Mechanism.
- To classify emotions into:
- Happy
- Sad
- Fear
- Anger
- Neutral
- To perform hyperparameter tuning for performance optimization.

CHAPTER 2

LITERATURE REVIEW

2.1 Video Emotion Recognition

Video Emotion Recognition has emerged as a significant research area because emotions are naturally conveyed through facial expressions and visual behavior.

Emotional information is represented through:

- Eyebrow movements
- Eye gaze
- Lip movements
- Facial muscle contractions
- Head movements

Recent advances in deep learning have significantly improved recognition performance.

2.2 Traditional Approaches

Early systems used handcrafted features such as:

- Facial landmarks
- Geometric facial measurements

- Action Units (AUs)

Machine learning classifiers included:

- SVM
- Random Forest
- Decision Tree
- Hidden Markov Models

These methods achieved moderate performance but failed to model complex emotional dynamics.

2.3 CNN-Based Emotion Recognition

Convolutional Neural Networks automatically learn spatial facial representations from image frames.

CNNs extract:

- Facial texture
- Facial structure
- Expression information
- Local facial features

However, CNNs cannot effectively capture temporal dependencies across video frames.

2.4 CNN-BiLSTM Emotion Recognition

CNN extracts spatial information while BiLSTM learns temporal emotional transitions.

Advantages:

- Spatial feature extraction
- Temporal sequence learning
- Better emotion recognition accuracy

2.5 Transformer-Based Approaches

Transformers utilize self-attention mechanisms to model contextual relationships across video frames.

Popular architectures include:

- BERT
- EmBERT
- RoBERTa

- Vision Transformers

These models capture long-range dependencies effectively.

2.6 Facial Landmark-Based Recognition

Facial landmarks provide geometric facial information.

Important facial regions:

- Eyebrows
- Eyes
- Nose
- Mouth
- Jawline

Eyebrow movement is particularly important for:

- Anger
- Fear
- Sadness
- Happiness

2.7 MELD Dataset

MELD (Multimodal EmotionLines Dataset) is derived from the TV series Friends.

Dataset contains:

- MP4 Videos
- Emotion Labels
- Conversational Dialogues

Selected Emotions:

- Happy
- Sad
- Fear
- Anger
- Neutral

2.8 Research Gap

Existing systems:

- Focus on multimodal inputs.
- Ignore eyebrow-specific features.
- Use limited temporal modeling.

Proposed system:

- Video-only emotion recognition.
- Facial landmark analysis.
- Eyebrow-aware feature extraction.
- CNN + BiLSTM + Attention architecture.
- Hyperparameter optimization.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Framework Overview

The proposed framework performs emotion recognition from video sequences using facial expressions and landmark information.

Workflow:

Input Video

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Frame Extraction

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Face Detection

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Face Alignment

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Facial Landmark Extraction

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CNN Feature Extraction

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BiLSTM Temporal Learning

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Attention Mechanism

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Softmax Classification

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Emotion Prediction

3.2 Video Preprocessing

Frame Extraction

Video clips are converted into sequential image frames.

Face Detection

Faces are detected using:

- OpenCV
- Dlib
- MediaPipe

Face Alignment

Faces are aligned to maintain consistency.

Face Cropping

Only facial regions are retained.

Image Resizing

Frame size:

224 × 224

Data Augmentation

- Rotation
- Horizontal Flip
- Brightness Adjustment
- Zoom

3.3 Facial Landmark Extraction

MediaPipe Face Mesh is used.

Extracted regions:

- Eyebrows
- Eyes
- Nose
- Mouth
- Jawline

Landmark coordinates:

(x, y, z)

These landmarks provide geometric emotional information.

3.4 CNN Spatial Feature Learning

CNN extracts:

- Facial textures
- Facial expressions
- Muscle activities
- Facial structures

Kernel Size:

3 × 3

Activation:

ReLU

Pooling:

Max Pooling

3.5 BiLSTM Temporal Modeling

BiLSTM captures emotional changes across video frames.

Advantages:

- Forward context learning
- Backward context learning
- Temporal dependency modeling

Hidden Units:

128

Layers:

2

3.6 Attention Mechanism

Attention identifies emotionally important frames.

Benefits:

- Focuses on relevant expressions.
- Improves classification performance.
- Reduces irrelevant information.

3.7 Proposed Model

Model Name:

EAEN

(Eyebrow-Aware Emotion Network)

Architecture:

Input Frames

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CNN

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BiLSTM

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Attention Layer

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Dense Layer

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Softmax

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Emotion Classification

3.8 Emotion Classification

Emotion Classes:

- Happy
- Sad
- Fear
- Anger
- Neutral

Softmax is used for final prediction.

3.9 Hyperparameter Configuration

Parameter	Value
Epochs	50
Batch Size	32
Learning Rate	0.0001
Kernel Size	3 × 3
Image Size	224 × 224
LSTM Units	128
Dropout	0.3
Optimizer	AdamW

3.10 Hyperparameter Tuning

Techniques:

- Grid Search
- Random Search
- Optuna

Tuning Parameters:

- Epochs
- Batch Size
- Learning Rate
- Kernel Size
- Hidden Units
- Dropout Rate

CHAPTER 4

RESULTS AND PERFORMANCE EVALUATION

Evaluation Metrics

Accuracy

Measures overall classification performance.

Precision

Measures correctness of positive predictions.

Recall

Measures ability to detect emotional classes.

F1-Score

Harmonic mean of Precision and Recall.

Weighted F1-Score

Handles class imbalance effectively.

Confusion Matrix

Shows class-wise prediction performance.

Expected Outcome

The proposed EAEN model is expected to:

- Improve emotion classification accuracy.
- Capture facial landmark dynamics.
- Utilize eyebrow movement information.
- Learn temporal emotional transitions.
- Achieve robust performance on MELD video data.

CONCLUSION

This project presents a Video Emotion Recognition framework using MELD dataset. The proposed Eyebrow-Aware Emotion Network (EAEN) combines facial landmark extraction, CNN-based spatial learning, BiLSTM temporal modeling, and attention mechanisms to classify emotions into Happy, Sad, Fear, Anger, and Neutral categories. The system aims to provide accurate and robust emotion recognition using only video information.